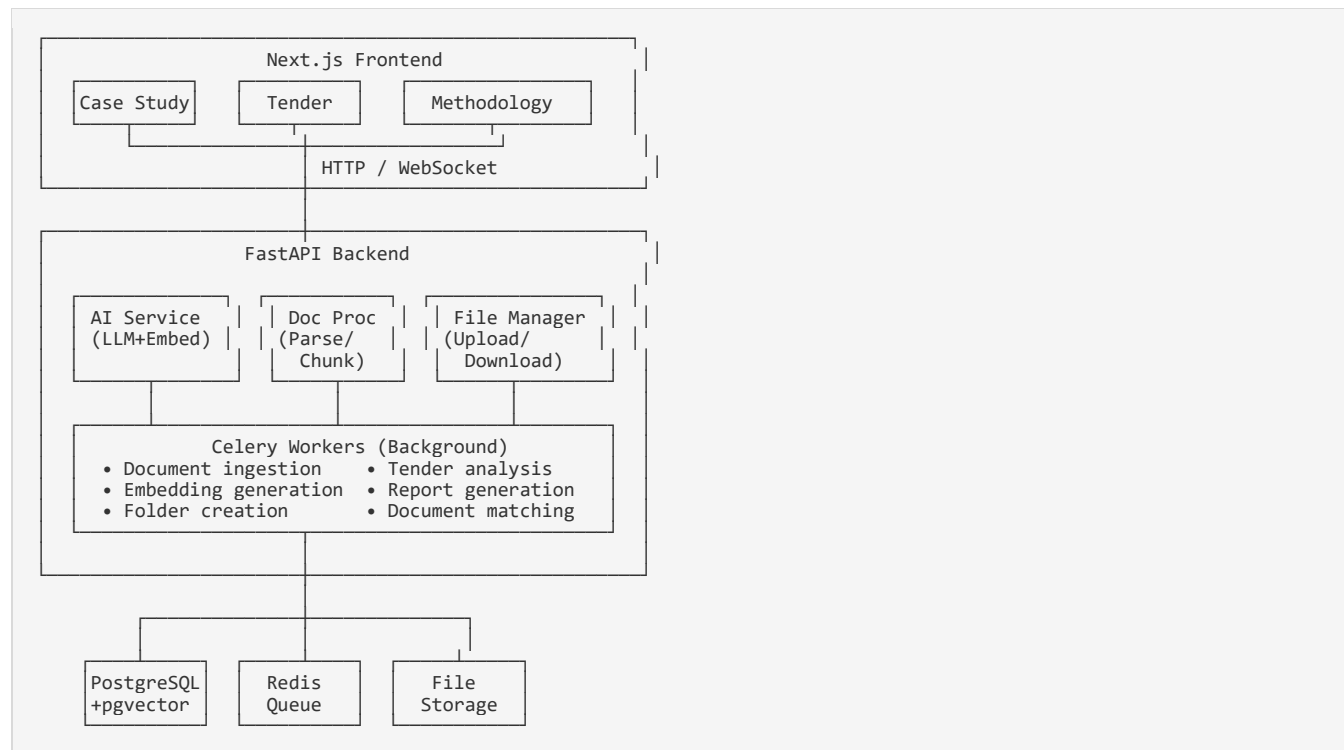


Implementation Plan

Architecture Overview

One unified platform. Three AI modules. Single backend. Single database.



Technology Stack

Backend

Layer	Technology	Purpose
API	FastAPI (Python 3.12)	REST endpoints, WebSocket, auth
Task Queue	Celery + Redis	Background processing (tender analysis, ingestion)
ORM	SQLAlchemy 2.0 + Alembic	Database models, migrations
Auth	JWT (python-jose + passlib)	Stateless authentication
Validation	Pydantic v2	Request/response schemas

AI

Layer	Technology	Purpose
LLM (light)	Gemini 2.0 Flash	Search, summaries, methodology queries
LLM (heavy)	Gemini 2.5 Pro	Tender requirement extraction
Embeddings	text-embedding-004 (Google)	Semantic vectors for all documents
PDF Parser	PyMuPDF (fitz)	Text + image extraction with page numbers
Word Parser	python-docx	.docx reading and report generation
PPT Parser	python-pptx	.pptx reading
OCR Fallback	Tesseract via pytesseract	Scanned PDF pages

Frontend

Layer	Technology	Purpose
Framework	Next.js 15 (App Router, TypeScript)	SSR, routing
Styling	Tailwind CSS 4 + shadcn/ui	Premium UI components
Icons	Lucide React	Consistent iconography
PDF Preview	react-pdf	In-browser document preview
Real-time	WebSocket (native)	Tender processing progress
Fonts	Inter (Google Fonts)	Modern typography

Storage

Layer	Technology	Purpose
Database	PostgreSQL 16 + pgvector	Metadata + vector search in one DB
Cache/Queue	Redis 7	Celery broker + response cache
Files	Local filesystem (structured)	All PDFs, images, docs

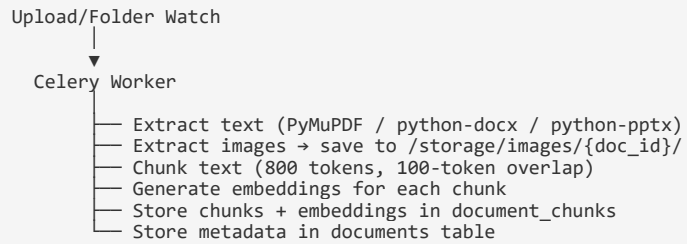
Infrastructure

Layer	Technology
Containers	Docker + Docker Compose
Reverse Proxy	Nginx
Process Monitor	Flower (Celery dashboard)

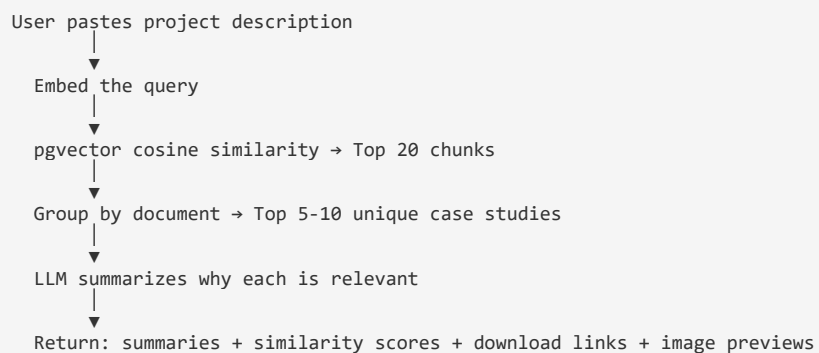
Module Designs

Module 1: Case Study RAG

Ingestion Flow



Search Flow



Module 2: Tender Analyzer

This is the complex module. It runs as a multi-step Celery pipeline with WebSocket progress updates.

Analysis Pipeline (8 steps)

```
Step 1: PARSE
  Upload tender PDF → PyMuPDF extracts text per page
  Track: page numbers, section headings, tables
  Fallback: Tesseract OCR for image-only pages

Step 2: CHUNK
  Split into 2000-token chunks with 200-token overlap
  Each chunk carries: page_start, page_end, section_heading

Step 3: EXTRACT
  For each chunk → LLM extracts requirements as JSON:
  {
    "requirement": "ISO 9001:2015 Certificate",
    "category": "certification",
    "is_mandatory": true,
    "page_number": 15,
    "section": "Eligibility Criteria",
    "deadline": null,
    "attachment_needed": true
  }

Step 4: MERGE
  Deduplicate requirements across chunks
  Resolve conflicts (same requirement from overlapping chunks)

Step 5: MATCH
  For each requirement where attachment_needed = true:
  → Search company documents by category + semantic similarity
  → Score: >0.85 = auto-match, 0.5-0.85 = suggest, <0.5 = missing

Step 6: REPORT
  Generate Word document (python-docx) containing:
  • Executive summary
  • Requirements table (requirement | page | mandatory | status)
  • Attachments checklist
  • Missing documents alert
  Also generate PDF version

Step 7: FOLDER
  Create folder: /storage/tenders/{tender_name}_{date}/
  Copy: tender PDF, generated report, all matched company docs

Step 8: REVIEW
  Mark analysis as "ready_for_review"
  User reviews matched documents → confirms or corrects
  On confirm → finalize folder
```

Progress Updates via WebSocket

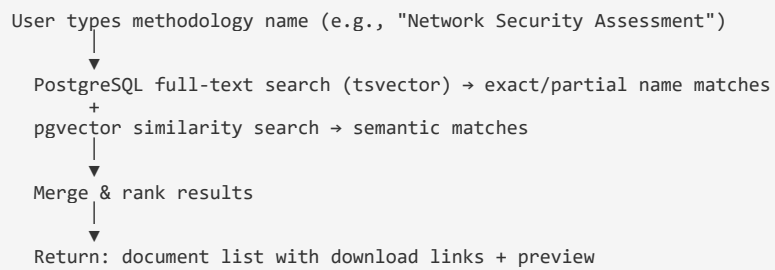
```
{ "step": 3, "total_steps": 8, "label": "Extracting requirements...", "progress": 37 }
{ "step": 5, "total_steps": 8, "label": "Matching company documents...", "progress": 62 }
{ "step": 8, "total_steps": 8, "label": "Ready for review", "progress": 100 }
```

IMPORTANT: The tender module includes a human review step (Step 8). The AI does not auto-finalize the folder the user must confirm that matched documents are correct. This prevents costly errors.

Module 3: Methodology Finder

The simplest module. No LLM needed for basic search — just full-text + vector search.

Search Flow



The user can:

- Search by name → instant results
- Browse all methodology docs in a list
- Download directly or preview in browser
- Click to open the file's location in the folder